

Phase 5 Project Development

FitBuddy AI Fitness Plan Generator

This phase records FitBuddy as a working local application. The core development result is a complete loop from profile entry to plan generation, storage, retrieval and feedback-based revision, supported by a browser interface and JSON API.

Technology and implementation layers

The project uses Python with FastAPI, Pydantic for request validation, SQLAlchemy and SQLite for local persistence, and Jinja templates with static CSS and JavaScript for the browser interface. Google's Gemini service is optional; the project also includes a local or demo generation path depending on configuration.

Implemented user capabilities

Capability	User-visible behavior
Profile capture	Collects goal and fitness context such as age, weight, intensity, experience and equipment; movement considerations are available in the supported form.
Weekly plan generation	Creates a seven-day schedule with workout guidance, recovery and a safety note.
Nutrition or recovery tip	Shows one general suggestion separately from the workout plan.
Plan persistence	Saves profile and plan information to the local SQLite database for later retrieval.
Feedback revision	Accepts a written request to adjust a saved plan and presents the latest version.
Operations and API	Provides local API documentation and an administrator-facing view where enabled by the project configuration.

Application workflow

The browser form submits a profile to the application. The application validates input, requests generated content, saves the profile and plan, and renders the result. Later feedback is associated with the saved profile or plan and produces a revision. The JSON API supports the same generation and retrieval concepts for integration or evaluation.

Development practices

- Keep API keys, administrator credentials and local settings in a private .env file rather than source control.
- Use the demo configuration for repeatable local development when live generation is not required.
- Keep database operations behind the persistence layer and validate request data at the application boundary.
- Review AI output for clarity, safe exercise guidance, recovery and fit with the user's stated equipment.

Phase outcome

The implementation supports FitBuddy's main value: a personalized, saved weekly plan that can be revisited and adjusted. Phase 6 checks this workflow across validation, generation, storage, feedback and administration.